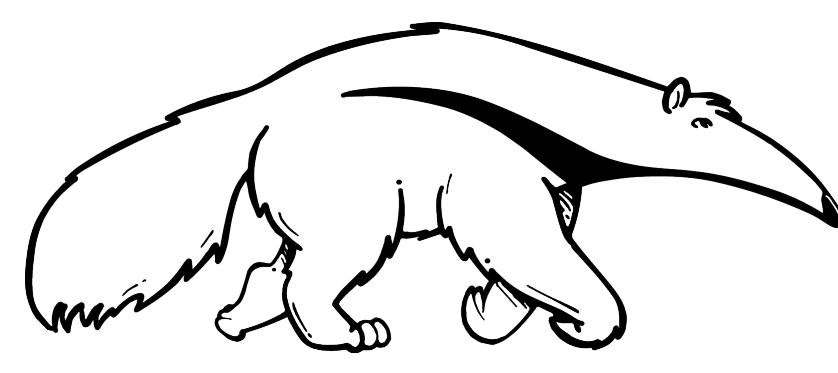
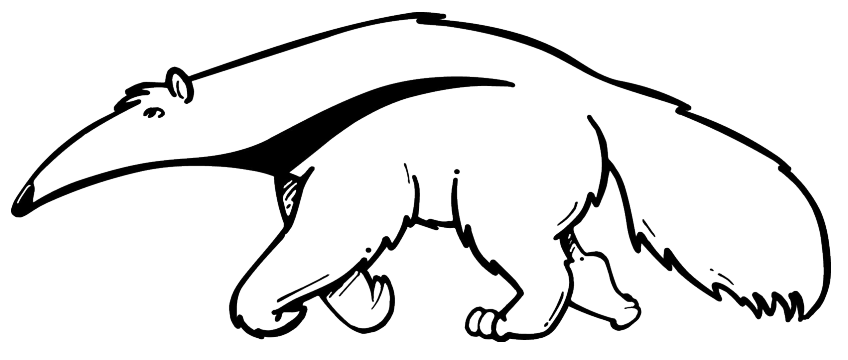




Development and Characterization of an In Vitro Ischemia-Reperfusion Model on the Heart-on-a-Chip Platform



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Introduction

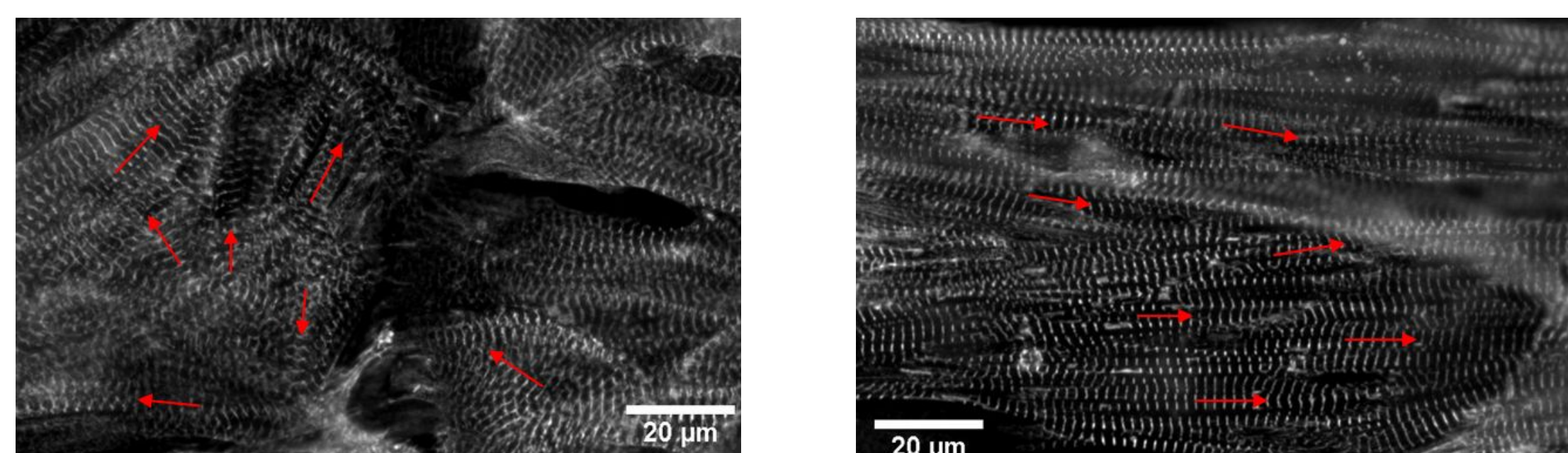
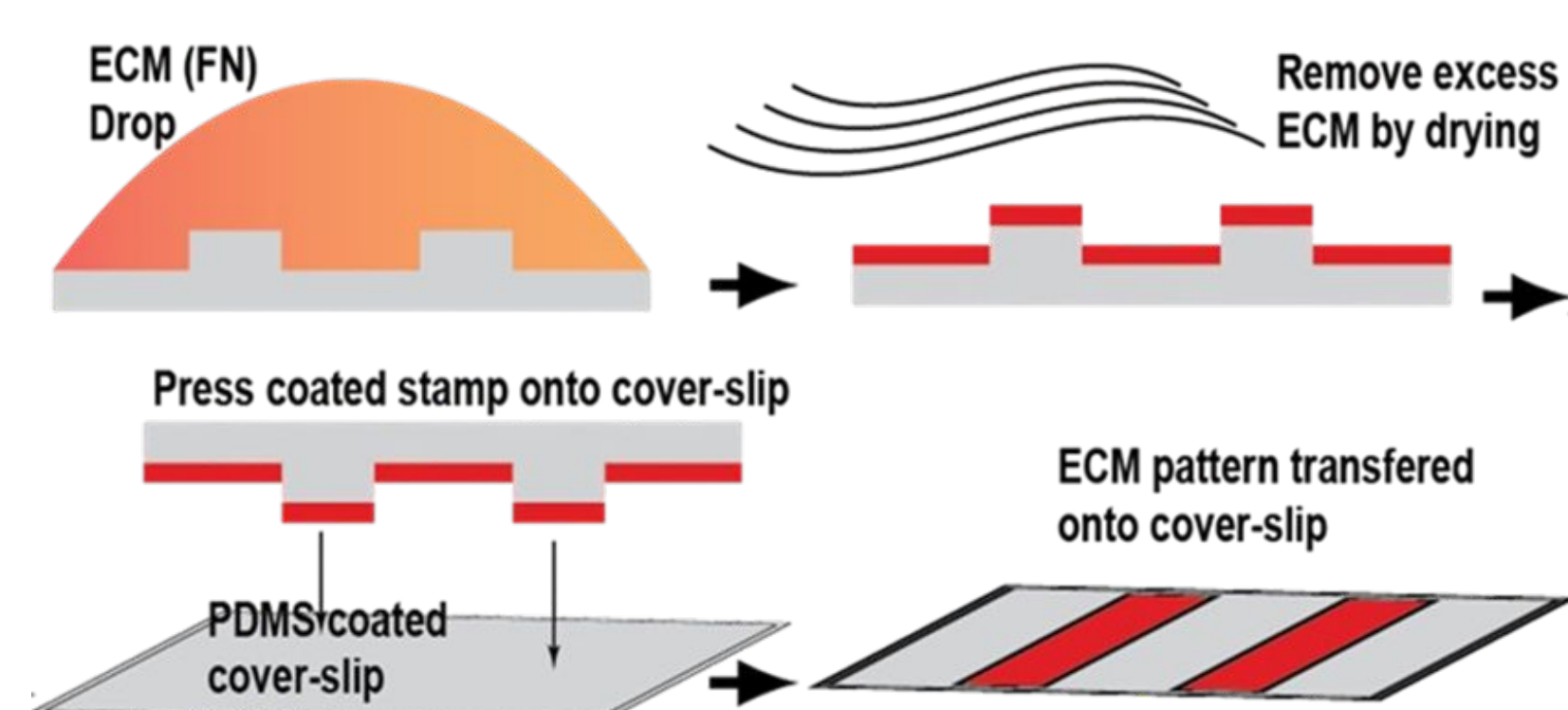
Myocardial infarction (MI) is a leading cause of death worldwide.

- Although the standard of care for MIs, is PCI, restoring blood flow or reperfusion itself can induce additional cardiac injury (I/R injury).
- Current in vivo models cannot readily isolate the cellular mechanisms underlying I/R injury.
- We developed a heart-on-a-chip platform to characterize the effects of I/R on cardiomyocyte viability and contractile function under controlled conditions.

Methods

Heart-on-a-chip Platform

Microcontact Printing

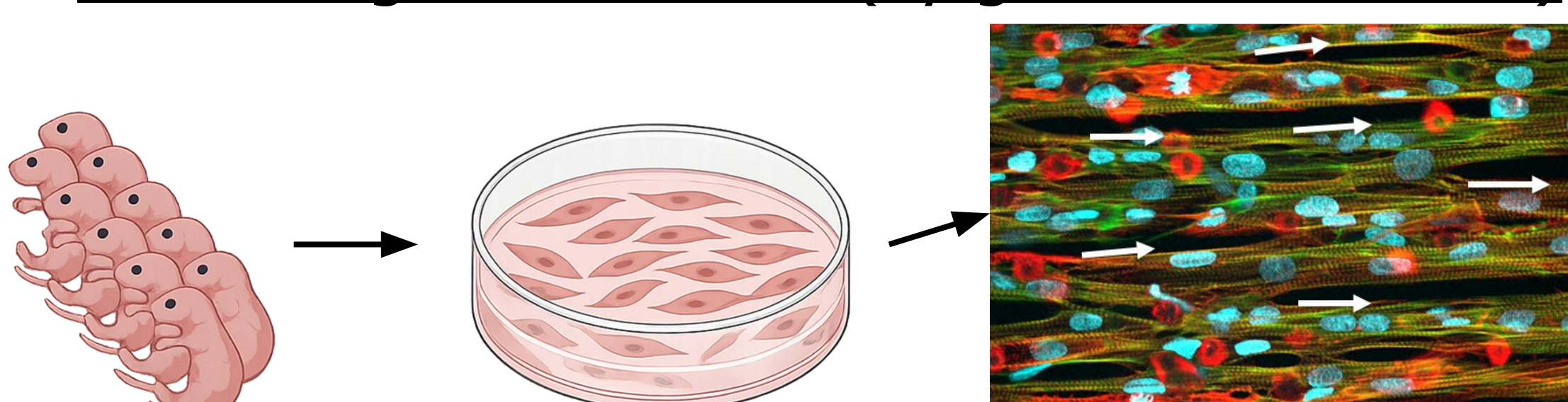


Isotropic sarcomeres

Anisotropic sarcomeres

→ Fibronectin patterning directs cells into uniaxially aligned that better recapitulate native myocardium

- Neonatal Rat Ventricular Myocytes (NRVMs)**
Harvesting and Isolation (by graduate students)



Ischemia - Reperfusion Protocol

Phase	Environment	Media	Duration
Ischemia	Gas chamber: 1% O ₂	serum-free, glucose-free DMEM	2 hours
Reperfusion	Standard incubator	M199 + 10% FBS	2 hours

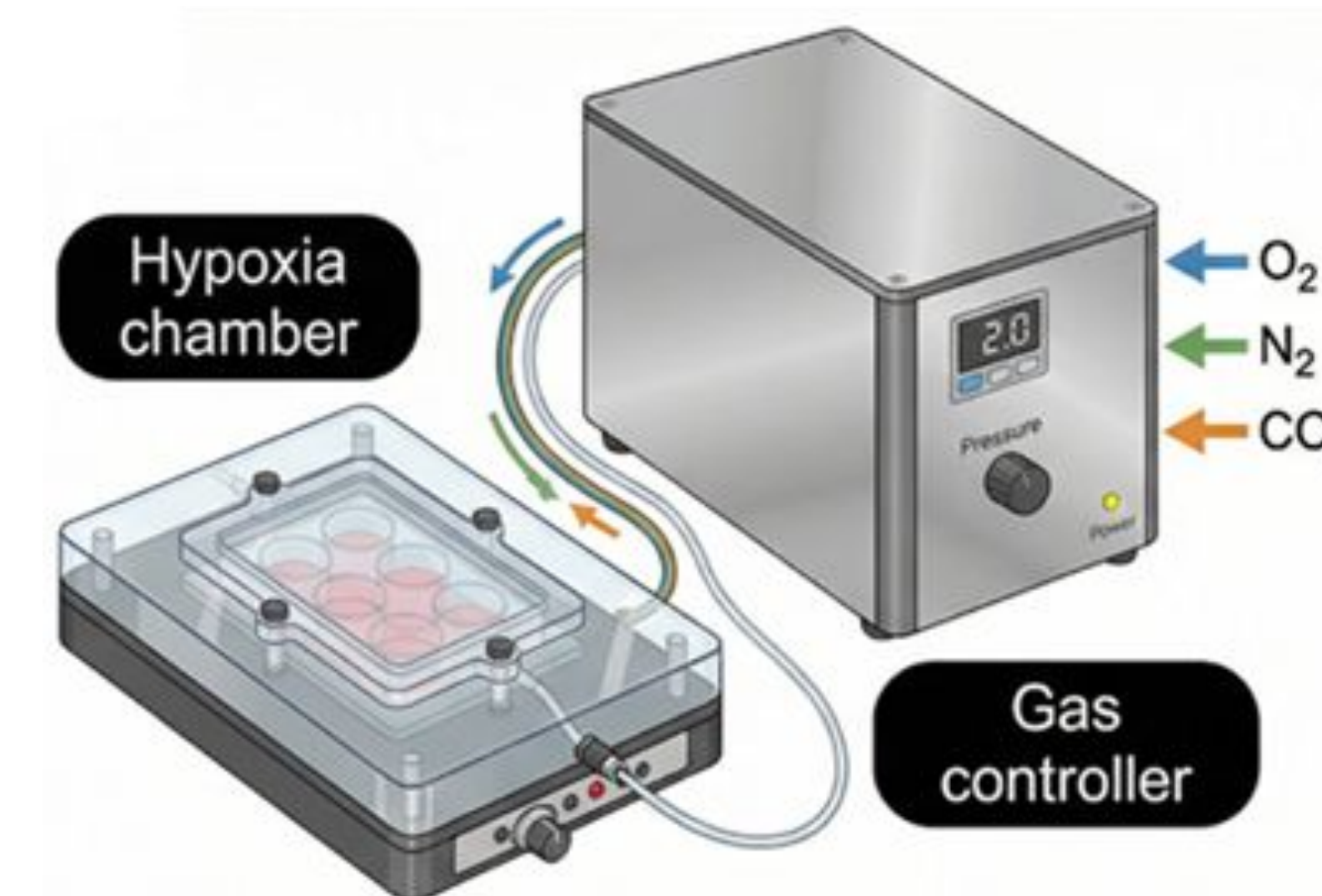


Figure 1. Gas Chamber Set-up

Preliminary Results

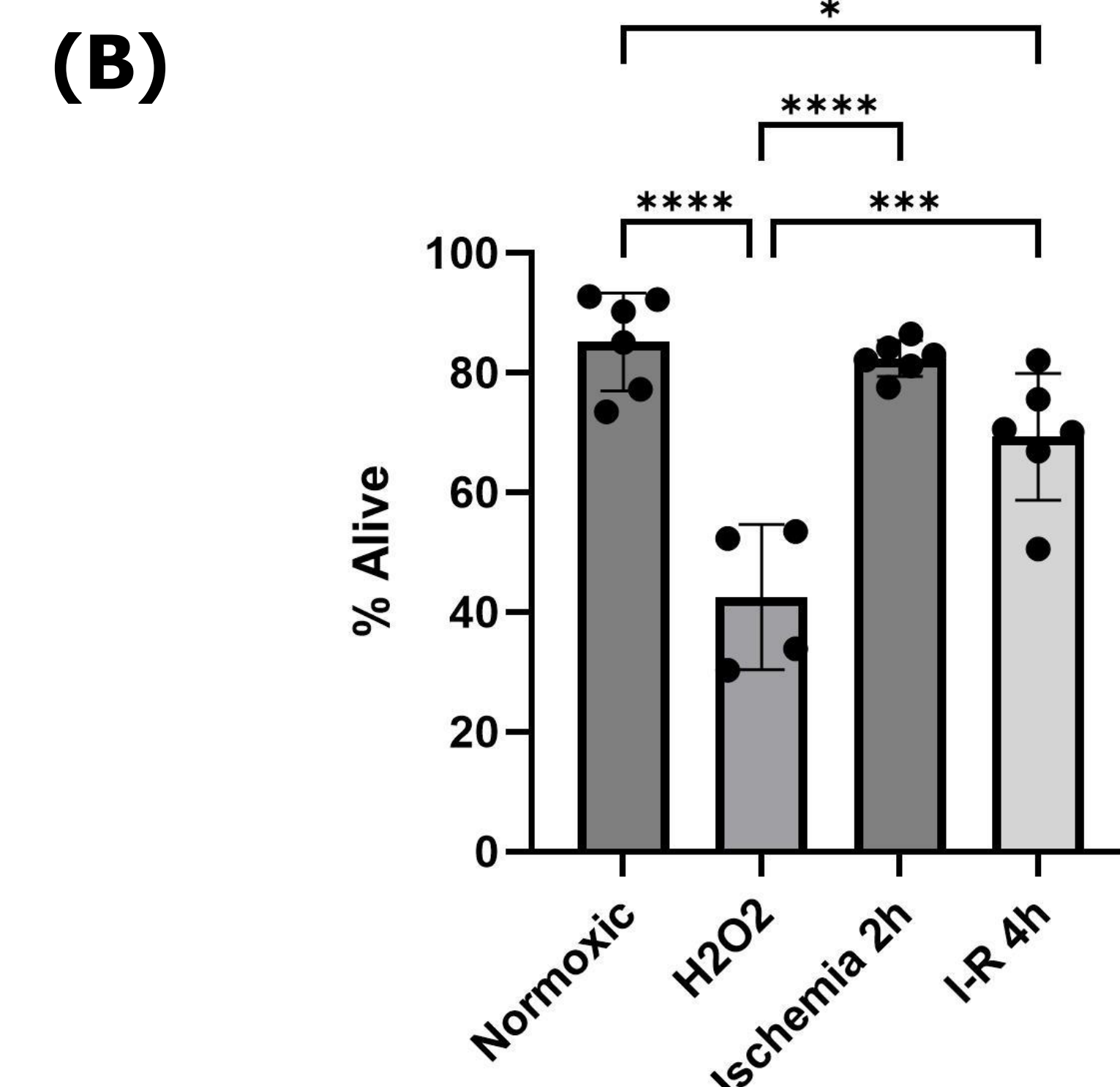
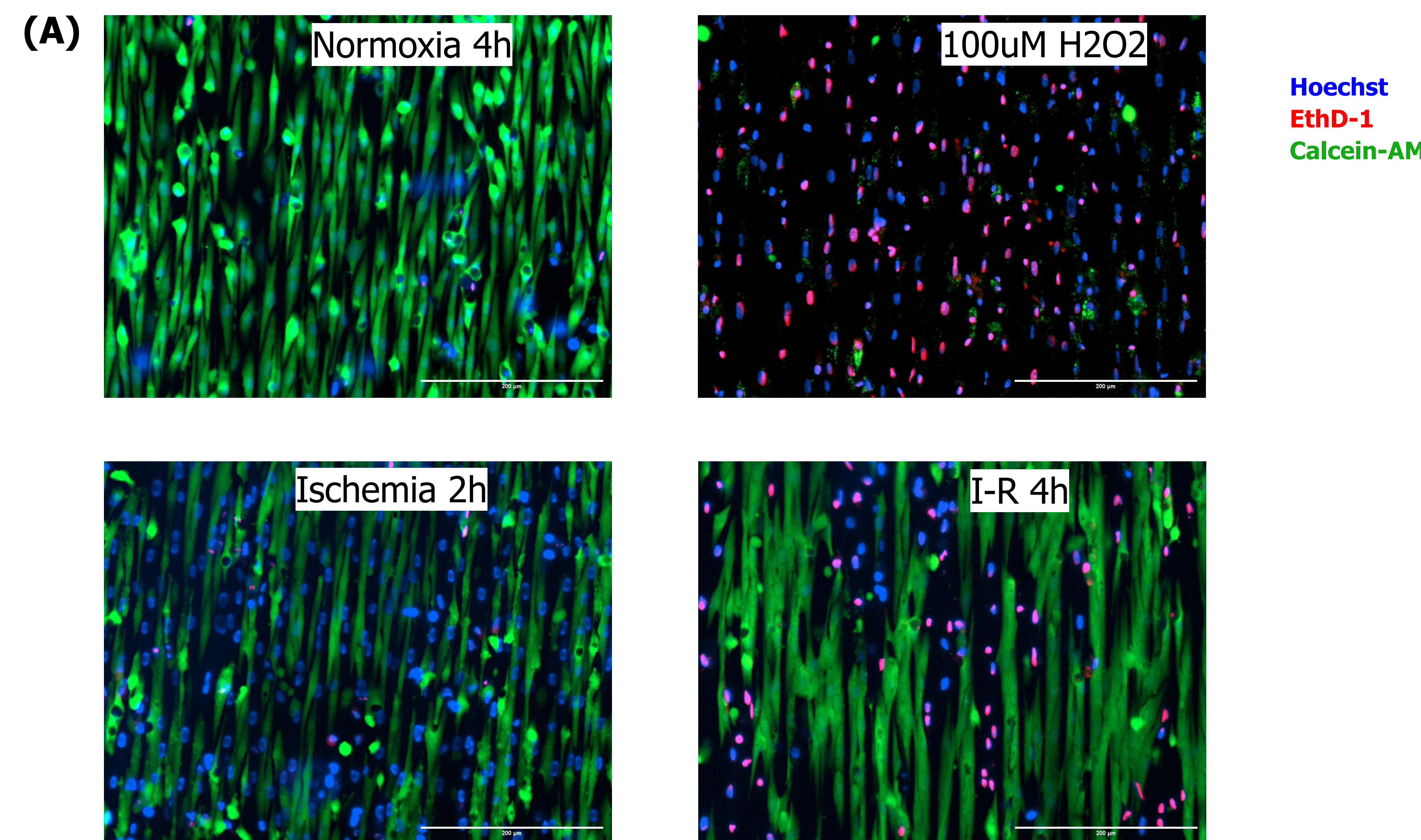


Figure 2. I/R injury reduces NRVM viability. (A) Representative fluorescence images. (B) Quantification of percent viable cells. * $p < 0.05$.

Discussion

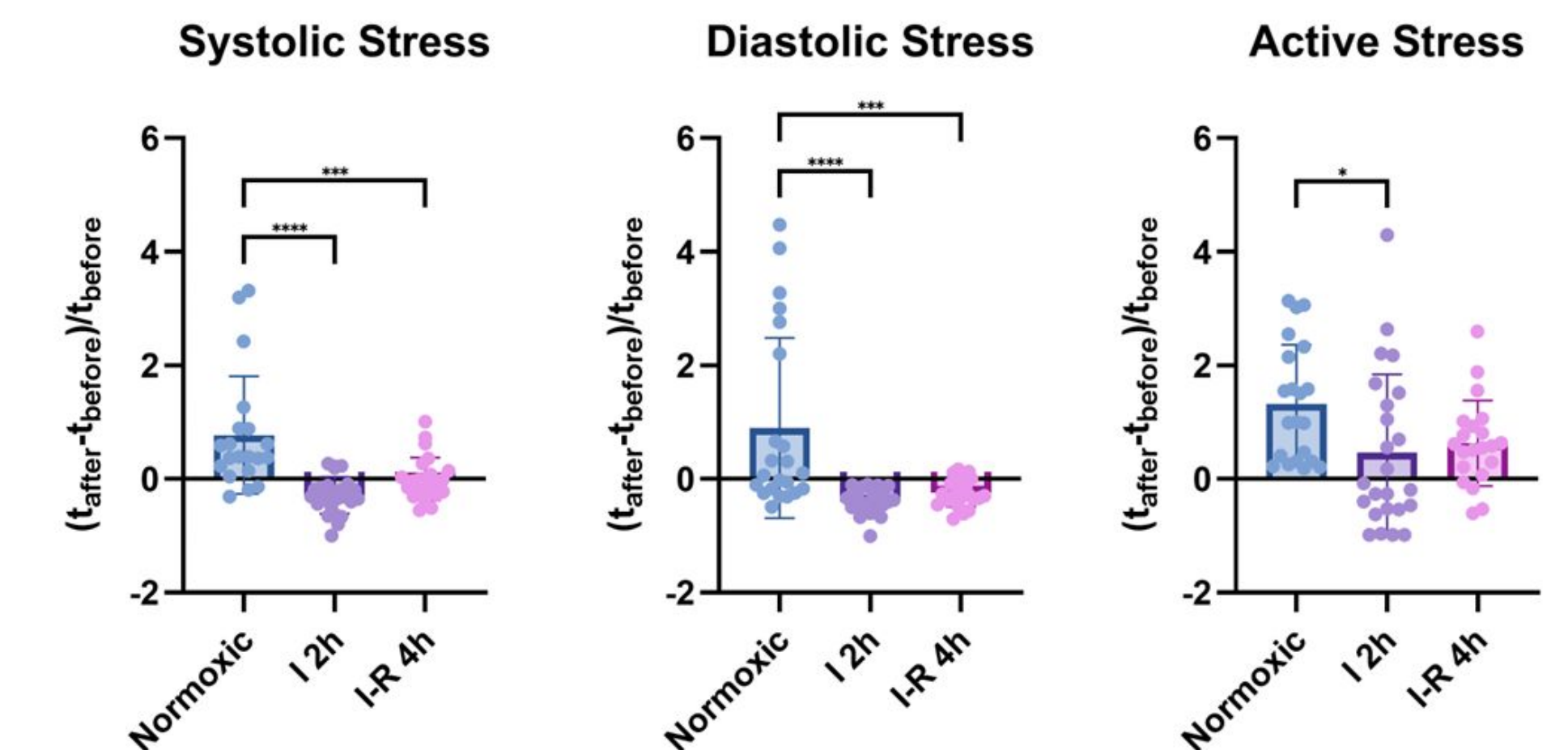


Figure 3. Cardiac contractility metrics. Normalized change in systolic, diastolic and active stress from baseline across conditions. * $p < 0.05$. Data collected by Khanh Tran

- Despite greater cell death in I-R 4h vs. Ischemia 2h, surviving cardiomyocytes demonstrated partial recovery of contractile function upon reperfusion, suggesting that restored oxygen and nutrient supply may rescue contractility in viable cells.

Future Direction

- Replicate viability findings across additional biological replicates to confirm reproducibility of the 2h/2h I/R protocol
- Expand functional and structural characterization using the MTF platform, including contractile stress and z-line OOP analysis
- Quantify cardiac troponin release via ELISA as a biochemical marker of I/R-induced injury
- Incorporate intermittent hypoxia preconditioning protocol prior to I/R insult to assess whether OSA-mimicking exposure attenuates cardiomyocyte injury

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Acknowledgements

